Implementation Plan

AstroIllustrator | Consolidated specification v2.1 | 22 October 2026

0. Repository and foundations

Create Api, Core, Infrastructure and Python renderer. Configure DI, typed options, SQLite migrations, renderer process adapter, job status, structured logs and local asset storage. Establish license inventory for star catalog, line sets and fonts. Gate: API can invoke Python CLI with a typed request and capture deterministic SVG output.

1. Orion scientific proof of concept

Select catalog, line-set definitions, projection and epoch. Render Orion with magnitude scaling, lines, labels and SVG/PNG. Compare to independent astronomy references; preserve manifest. Gate: Orion coordinates and line topology validated; output regenerates byte-equivalently or with documented stable geometry hashes.

2. Generalized sky scenes

Introduce SkyChartRequest and common layers. Add MultipleConstellations and SummerTriangle definitions with Vega/Deneb/Altair. Add auto-framing and consistent presets. Gate: multiple constellations coexist without duplicate stars or incorrect connecting edges.

3. Observer-dependent horizon and seasonal scenes

Add observer, UTC time, Astropy AltAz transforms, horizon clipping, azimuth-centered view, projections and orientation tests. Generate southern summer horizon from Prishtina. Gate: positions agree within documented tolerance with an independent reference for several dates/locations; horizon direction is correct.

4. Scientific diagram engine

Build shared geometry primitives: axes, rays, angle arcs, labels, sphere outlines and projections. Implement altitude/azimuth, zenith, Polaris/latitude, celestial sphere and angular separation templates. Gate: numerical and visual tests prove 45 degree angle is drawn as 45 degrees; computed Polaris altitude is not confused with NCP altitude.

5. Atlas authoring and batch exports

Add chapter/figure manifest, language dictionaries, print profiles, SVG/vector PDF, high-resolution PNG, batch rendering and reproducibility metadata. Gate: regenerate 10 diverse figures from manifests with no geometry drift; print-size legibility passes editorial review.

6. AI prompt-to-illustration

Add IIllustrationPromptInterpreter behind provider adapter, structured output schema, strict validation, ambiguity handling and manual fallback. First prompts: “Polaris altitude from Prishtina” and “Summer Triangle with cyan lines”. Gate: test prompts map to valid typed requests; model never supplies authoritative angles/coordinates; no AI dependency for manual path.

7. Optional AI scenery

Generate star-free landscapes, track provider rights/cost, composite as separate layer, validate scientific overlay geometry and label illustrative scene. Gate: star positions and angles remain unchanged after composition; asset provenance recorded.

8. Optional post-MVP extensions

RAG educational captions with curated references, interactive diagram controls, more orbital templates, UI and tool-calling assistant only if real author workflow warrants them. Avoid multi-agent systems, microservices, CQRS, Kafka, Kubernetes and vector databases without evidence of need.

9. Delivery definition of done

Each feature includes typed contract, validation, tests, source/license metadata, documentation, deterministic output, failure behavior and at least one atlas-ready sample. Recommended first two demonstration outputs: Orion star chart and Polaris altitude diagram; together they prove the two core scientific engines. Document control: This version supersedes earlier 1.0 and 1.1 scope descriptions. Keep requests, manifests, diagrams and AI integrations aligned with this specification.

v2.1 additions: global observer support, time-aware conjunctions and AstroIllustrator naming

Revised milestones

0. AstroIllustrator solution, API/Core/Infrastructure, Python renderer and SQLite. 1. Orion verified star chart. 2. Generalized sky scenes and asterisms. 3. Global ObserverLocation, UTC/local time conversion, horizon views. 4. Scientific diagrams (Polaris altitude/zenith). 5. Solar system ephemerides, sunset/twilight and fixed-date conjunction render. 6. Event search within date ranges and visibility ranking. 7. AI structured prompt-to-request interpretation. 8. Optional artistic composites, batch publishing and RAG.

Event implementation test gates

Golden tests against independent ephemeris reference for Sun, Moon and planets at diverse latitudes and seasons; polar day/no sunset; DST boundaries; southern hemisphere; west azimuth and below-horizon filtering; date range closest-approach search. Render regression tests compare geometry and manifests, not arbitrary image pixel identity.

Vertical slice: western sky

From a supplied Rome date and solar-relative time: calculate sunset; determine positions of Venus and Jupiter; report actual separation/visibility; render western twilight scene; export SVG/PNG; persist reproducibility manifest. Do not make conjunction claim unless numerical criteria are met.

Scientific note: illustrative dates and prompts are examples, not verified event predictions. All published charts require computed ephemerides and editorial review.